

Peter A. Lilley

Software Engineer - Local LLM Inference, Language-to-Geometry Systems, Production Tooling

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SUMMARY

I am a software engineer. I build production systems that other professionals use daily. I built a language-to-geometry pipeline. It uses a plain-language interface to generate 3D models and set up engineering deliverables. I use locally hosted LLM inference to handle intent. I use deterministic C# and .NET code to construct and validate geometry. I also build OCR ingestion and automation tools. These tools cut repetitive drafting from 8 to 12 hours to about 30 seconds per cycle. I work in C#, Python, C++17, TypeScript, and Dart.

SKILLS

Languages: C#, Python, C++, Dart, TypeScript, JavaScript, SQL, LISP, VBA

AI and local models: Ollama, LM Studio, Open WebUI, Google APIs, OCR, speech-to-text, prompt engineering, text matching

Databases and Data Systems: SQLite, Firebase, Postgres, XData schemas, data modeling, JSON structuring, graph calculators

Engineering and math: Matrix math, 3D graphics, Express, React, Git, CMake

Domain systems and quality: CAD plugins, Revit automation, 3D metadata, ARCore, part dimensioning, coordinate measuring machines, ISO 9001

EXPERIENCE

David Mason & Associates - Staff Technician (CAD automation / Civil 3D tools)

Jul 2024 - Present

St. Louis, MO

- I build CAD generation tools. These turn natural-language input into 3D models and drawing setups using local LLM inference and deterministic C#/.NET. I introduced this capability at the firm.
- I built OCR ingestion software. It parses existing PDF plan sets to auto-check and auto-populate them. This replaced a manual mark-up loop.
- I automate repetitive drafting with Python and Dynamo. This cuts cycles from 8 to 12 hours to about 30 seconds. It reduces errors before licensed-engineer review by about 25%.
- I design typed metadata schemas. These attach to model entities. I coordinate standards automation across civil, mechanical, plumbing, and architectural disciplines.
- I built C# custom commands for Civil 3D to automate layout, rename title blocks, audit drawings, and fix broken data. They rename up to 40 layouts simultaneously in under 1.7 seconds. This automation eliminates 15-to-20-second manual loading delays per layout during repetitive series renames.
- I used C# to parse messy survey data into Civil 3D XData, building graph lookup tables to map 3D structures to exact flow lines.

Component Bar Products - Quality Engineer - Manufacturing / Precision Machining

Apr 2023 - Jun 2024

O'Fallon, MO

- I programmed and operated coordinate measuring machines. I verified precision-machined parts to +/-0.001 inch or tighter against GD&T specifications.
- I validated dimensional tolerances against CAD models and customer prints. Machining problems showed as geometry mismatches, not downstream scrap.
- I enforced ISO 9001 across production lines. I integrated Production Part Approval Process documentation. I managed calibration and traceability of metrology assets.

Heideman & Associates, Inc. - Revit/CAD Technician - Mechanical/Plumbing

May 2018 - Apr 2019

Fenton, MO

- I produced mechanical and plumbing construction documents. I built as-built models in Revit and AutoCAD.
- I audited standards for cross-discipline consistency.
- I built automated data libraries for building-system objects. I standardized hundreds of outdated details.

PROJECTS

Digital Twin Pro

Personal product, Free Parameter LLC | Flutter/Dart 3, SQLite, Firebase, Google Gemini API, ARCore | github.com/freeParameterized/digital-twin-pro

- I designed, funded, and shipped a cross-platform 3D inventory application. I released it to a Google Play beta and built a Windows desktop version.
- I wrote the 3D renderer by hand instead of using a game engine. I used a painter's-algorithm scene rendering with a Z-sorted draw queue and custom projection.
- I added photo-based item detection and voice entry via a paid Gemini API.
- I built the database with SQLite to store inventory data locally. I included a move audit log. I added QR booklets and CSV/JSON export.

CAD integration bridge

C++17, Dear ImGui, Windows COM | local repository

- I built a C++17 Dear ImGui host. It COM-bridges BricsCAD, AutoCAD, and Civil 3D.
- It discovers installed SDKs at runtime. It hot-loads LISP, C#, and BRX plugins. It has about 14,000 lines of code.

Interactive Portfolio (this site)

React, TypeScript, Three.js, Express, Ollama

- I built a React Three Fiber portfolio over an Express API.
- It retrieves data from a curated corpus. It answers questions through a local model.
- It uses on-device speech. It degrades to extractive answers when no model runs.

EDUCATION

Calculus II (CLEP Credit) - Saint Louis University (SLU)

2018

St. Louis, MO

Accepted by Ranken Technical College for their calculus curriculum.

Coursework toward Latin American Studies with Technical Applications - Missouri University of Science and Technology

2019 - 2020

Rolla, MO

Coursework including rigorous, no-calculator College Algebra (determinants, Cramer's rule, introductory linear algebra concepts) and general education. Technical fluency in Spanish.

A.A.S., Building Systems Engineering Technology - Ranken Technical College

2019

Wentzville, MO